

Experiment: 9

Massive MIMO Capacity vs SNR and Gain Using MATLAB

Aim

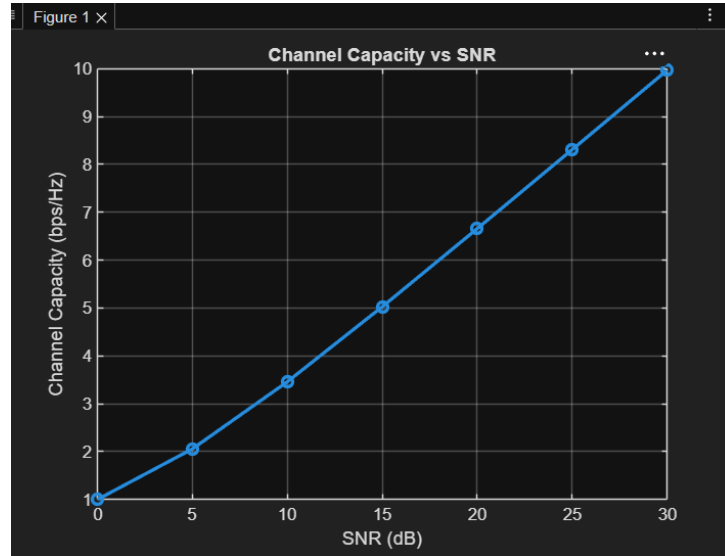
To analyze the performance of a Massive MIMO system by studying channel capacity, signal-to-noise ratio (SNR), and antenna gain using MATLAB.

Objective – 1 Plot of Channel Capacity (bps/Hz) versus SNR (dB).

MATLAB Code:

```
clc;
clear;
close all;
snr = 0:5:30;
snr_linear = 10.^(snr/10);
capacity = log2(1 + snr_linear);
figure;
plot(snr,capacity,'-o','LineWidth',2,'MarkerSize',6);
grid on;
xlabel('SNR (dB)');
ylabel('Channel Capacity (bps/Hz)');
title('Channel Capacity vs SNR');
```

Expected Output



Objective – 2 Bar graph comparing SNR values under different communication scenarios.

Analyze the channel capacity of a Massive MIMO system for different signal-to-noise ratio

(SNR) values using MATLAB. Observe how channel capacity increases with improving signal

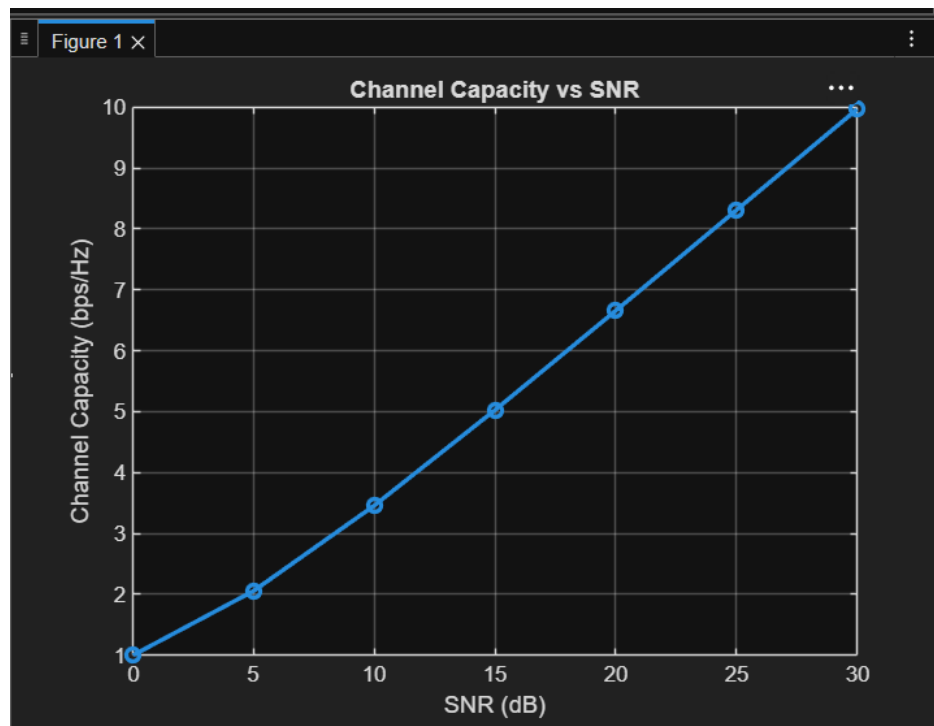
quality based on the Shannon capacity principle.

MATLAB Code:

```
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snr = 0:5:30;
snr_linear = 10.^(snr/10);
capacity = log2(1 + snr_linear);
figure;
```

```
plot(snr, capacity, '-o', 'LineWidth', 2, 'MarkerSize', 6);  
grid on;  
xlabel('SNR (dB)');  
ylabel('Channel Capacity (bps/Hz)');  
title('Channel Capacity vs SNR');
```

Expected Output



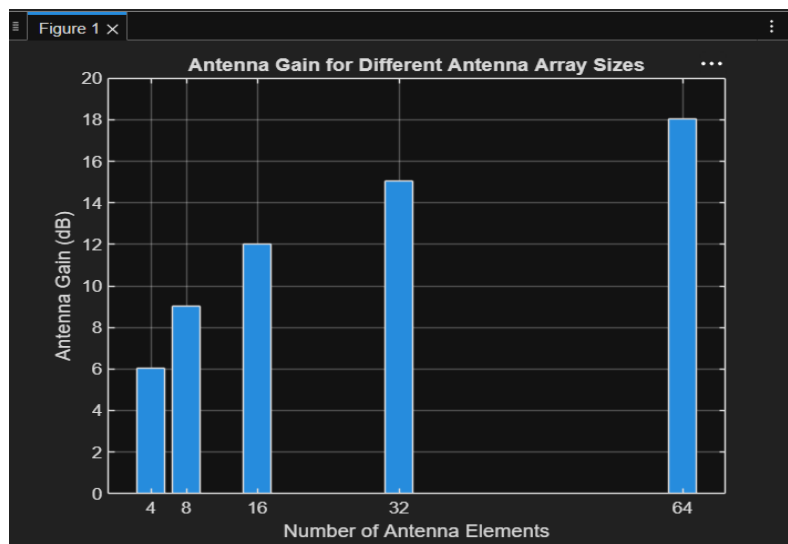
Objective – 3 Bar graph showing Antenna Gain for different antenna array sizes.

MATLAB Code:

```
clc;  
clear;  
close all;
```

```
antennas = [4 8 16 32 64]; % Number of Antenna Elements gain =  
10*log10(antennas); % Antenna Gain (dB)  
  
figure  
  
bar(antennas, gain)  
  
grid on  
  
xlabel('Number of Antenna Elements')  
  
ylabel('Antenna Gain (dB)')  
  
title('Antenna Gain for Different Antenna Array Sizes')
```

Expected Output



Result

The MATLAB analysis successfully demonstrated the relationship between channel capacity, signal-to-noise ratio, and antenna gain in a Massive MIMO system. The results showed that higher SNR and larger antenna arrays improve communication performance by increasing channel capacity and directional gain.